

VNUHCM - University of Science
Faculty of Information Technology

CS486 – Lecture 4

Query Languages

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Content

- Introduction
- Define relation schemas
- Manipulate on relation schemas
 - Data modification
 - Data retrieval
 - Basic query
 - * On one schema
 - * **On two or many schemas**
 - Advanced query
- Summary

In relational algebra

- Selection
- Projection
- **Cartesian product**
- **Join operation**
- Set operations
- Division operation
- Other operations
- Modification operations

Cartesian product

- Cross-product
 - Is used to combine tuples from two relations in a combinatorial fashion
- Denotation $R \times S$
- The result is a relation Q
 - Q has one tuple for each combination of tuples, one from R and one from S
 - If R has u tuples and S has v tuples,
 - Then Q will have $(u \times v)$ tuples
 - If R has n attributes and S has m attributes,
 - Then Q will have $(n + m)$ attributes ($R^+ \cap S^+ = \emptyset$)

Cartesian product

■ Example

R	A	B
	α	1
	β	2

S	X	C	D
	α	10	+
	β	10	+
	β	20	-
	γ	10	-

$\rho_{(X,C,D)}(S)$

$R \times S$

A	B	X	C	D
α	1	α	10	+
α	1	β	10	+
α	1	β	20	-
α	1	γ	10	-
β	2	α	10	+
β	2	β	10	+
β	2	β	20	-
β	2	γ	10	-

Cartesian product

■ Example

R	A	B
	α	1
	β	2

S	B	C	D
	α	10	+
	β	10	+
	β	20	-
	γ	10	-

R $\times$ S	A	R.B	S.B	C	D
	α	1	α	10	+
	α	1	β	10	+
	α	1	β	20	-
	α	1	γ	10	-
	β	2	α	10	+
	β	2	β	10	+
	β	2	β	20	-
	β	2	γ	10	-

unambiguous

Cartesian product

- Cartesian product is often followed by a selection operation

$$R \times S$$

A	R.B	S.B	C	D
α	1	α	10	+
α	1	β	10	+
α	1	β	20	-
α	1	γ	10	-
β	2	α	10	+
β	2	β	10	+
β	2	β	20	-
β	2	γ	10	-

$$\sigma_{A=S.B}(R \times S)$$

A	R.B	S.B	C	D
α	1	α	10	+
β	2	β	10	+
β	2	β	20	-

Example 6

- For each department, list out the information of the manager

DName	DNumber	MgrSSN	MgrStartDate						
Research	5	333445555	05/22/1988						
Administration	4	987654321	01/01/1995						
DName	DNumber	MgrSSN	MgrStartDate	SSN	FName	LName	...		
Headquarters	1	888665555	06/19/1981						
Research	5	333445555	05/22/1988	333445555	Franklin	Wong			
Administration	4	987654321	01/01/1995	987987987	Jennifer	Wallace			
DName	FName	Minit	LName	BDate	Address	Sex	Salary	SuperSSN	DNo
Headquarters	John	1	B	888665555	01/01/1981	7318866555	Male	James	333445555.. 5
333445555	Franklin	T	Wong	12/08/1955	638 Voss	Male	40000	888665555	5
999887777	Alicia	J	Zelaya	01/19/1968	3321 Castle	Female	25000	987654321	4
987654321	Jennifer	S	Wallace	06/20/1941	291 Berry	Female	43000	888665555	4
666884444	Ramesh	K	Narayan	09/15/1962	975 Fire Oak	Male	38000	333445555	5
453453453	Joyce	A	English	07/31/1972	5631 Rice	Female	25000	333445555	5
987987987	Ahmad	V	Jabbar	03/29/1969	980 Dallas	Male	25000	987654321	4

Example 6

- Step 1:
 - Cartesian product DEPARTMENT and EMPLOYEE
- Step 2:
 - Select tuples that satisfy the condition $\text{MgrSSN} = \text{SSN}$

Example 7

- Find the highest salary in company

FName	LName	...	Salary	...	...	Salary	...
John	Smith	...	30000	...	...	30000	...
Franklin	Wong	...	40000	...	...	40000	...
Alicia	Zelaya	...	25000	...	...	25000	...
Jennifer	Wallace	...	43000	...	...	43000	...



Example 7

- Step 1:
 - Select salaries which are not the highest one

- Step 2:
 - Let do the difference of the set of salary and salary in $R3$

Example 8

- Find the departments that have the same locations as the department 5

Which location does the department 5 have?

DNumber	DLocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

Which department will have locations which are in that set

DNumber	DLocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

Example 8

- Step 1:
 - Find the locations of the department 5

- Step 2:
 - Select the departments that have the same locations as department 5's location

Join operation

- Is used to combine related tuples from 2 relations into single tuples
- Denotation $R \bowtie S$
 - $R(A_1, A_2, \dots, A_n)$ and $S(B_1, B_2, \dots, B_m)$
- Result is a relation Q
 - Has $(n + m)$ attributes $Q(A_1, A_2, \dots, A_n, B_1, B_2, \dots, B_m)$
 - A tuple of Q is a combination of tuples from R and S satisfying some join condition
 - The form : $A_i \theta B_j$
 - A_i : the attribute from R , B_j : the attribute from S
 - A_i and B_j have the same domain
 - θ : comparison operators $\neq, =, <, >, \leq, \geq$

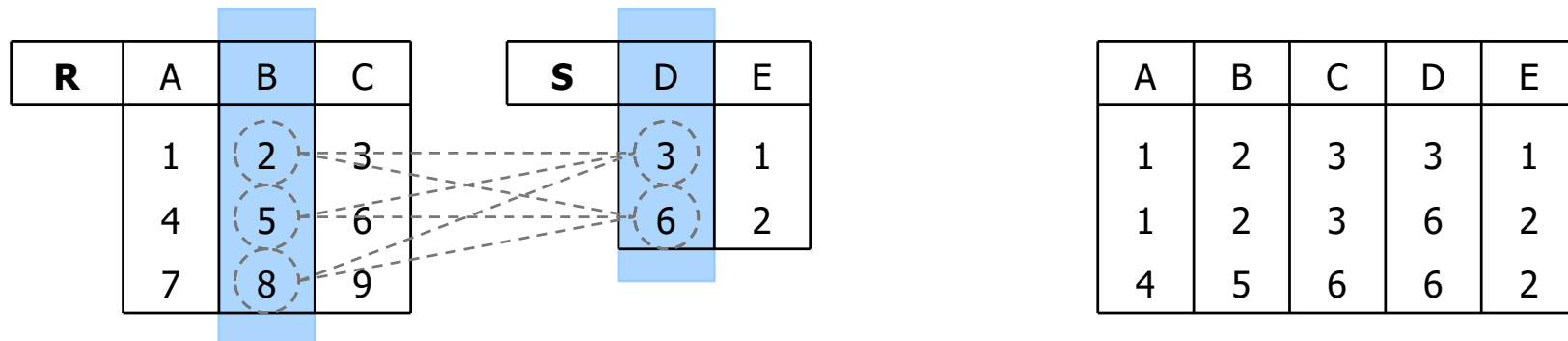
Join operation

■ Categories

- **Theta join** pairs tuples using one specific condition
 - Denotation $R \bowtie_C S$
 - C refers to an arbitrary condition for attributes
- **Equijoin** when C involves equality comparisons only
- **Natural join**
 - Denote $R \bowtie S$ or $R * S$
 - $R^+ \cap S^+ \neq \emptyset$
 - Only one join attribute is kept

Join operation

- Example of theta join



$$R \bowtie_c S = \sigma_c(R \times S)$$

Join operation

■ Example of equijoin

R	A	B	C
	1	2	3
	4	5	6
	7	8	9

S	D	E
	3	1
	6	2

$$R \bowtie_{C=D} S$$

A	B	C	D	E
1	2	3	3	1
4	5	6	6	2

R	A	B	C
	1	2	3
	4	5	6
	7	8	9

S	S.C	D
	3	1
	6	2

$$R \bowtie_{C=S.C} S$$

A	B	C	S.C	D
1	2	3	3	1
4	5	6	6	2

$$\rho_{(S.C,D)} S$$

Join operation

- Example of natural join

R	A	B	C
	1	2	3
	4	5	6
	7	8	9

S	C	D
	3	1
	6	2

$R \bowtie S$

A	A	B	B	C	C	S	C	D	D
1	1	2	2	3	3	3	1	1	1
4	4	5	5	6	6	6	2	2	2

Example 9

- Find the employee whose salary is greater than the salary of the employee “John”

EMPLOYEE(LName, Minit, FName, SSN, ..., **Salary**, DNo)

Example 10

- For each employee, find the information of the department where he/she is working for

EMPLOYEE(LName, FName, SSN, ..., **DNo**)

DEPARTMENT(DName, **DNumber**, MgrSSN, MgrStartDate)

Example 11

- Find the locations for each department

DEPARTMENT(DName, **DNumber**, MgrSSN, MgrStartDate)

DEPT_LOCATIONS(**DNumber**, DLocation)

Example 12

- Find the SSN and first name of employees who work for the department “Research”

Example 12 – SQL

- Find the SSN and first name of employees who work for the department “Research”

FROM clause

Unspecified WHERE clause

```
SELECT SSN, DNumber
FROM   EMPLOYEE, DEPARTMENT
WHERE  TRUE
```

SSN	DNumber
333445555	1
333445555	4
333445555	5
987987987	1
987987987	4
987987987	5
...	...

FROM clause

Alias name

```
SELECT DName, DLocation  
FROM DEPARTMENT, DEPT_LOCATIONS  
WHERE DNumber=DNumber
```

```
SELECT DName, DLocation  
FROM DEPARTMENT AS DE, DEPT_LOCATIONS AS DL  
WHERE DE.DNumber=DL.DNumber
```

FROM clause

Alias name

```
SELECT FName, BDate, Dependent_Name, BDate  
FROM EMPLOYEE, DEPENDENT  
WHERE SSN=ESSN
```

```
SELECT FName, EM.BDate, Dependent_Name, DE.BDate  
FROM EMPLOYEE EM, DEPENDENT DE  
WHERE SSN=ESSN
```

ORDER BY clause

- Is used for presenting a query in sorted order
- Syntax

```
SELECT <List_of_columns>  
FROM <List_of_tables>  
WHERE <Conditions>  
ORDER BY <List_of columns>
```

- ASC (default)
- DESC

Example

```
SELECT ESSN, PNo
FROM   WORKS_ON
ORDER BY ESSN DESC, PNo
```

ESSN	PNo
999887777	10
999887777	30
987987987	10
987987987	30
987654321	10
987654321	20
987654321	30

Example 13

- For each project locating in “Stafford”, find its number, the department number that controls this project, the last and first name of the manager, as well as his/her birth date and address

Example 14

- Find the last and first name of employees who work for the department 5 and work on the project “Product X” with working hours are larger than 10

Example 15

- Find the first and last name of employees and their supervisor

Example 16

- Find the last and first name of employees who are supervised directly by “Franklin Wong”

Example 12 – Relational calculus

- Find employees (SSN) who work for the department “Research”

$t.SSN \mid t \in \text{EMPLOYEE}$

$s \in \text{DEPARTMENT} \wedge s.DNAME = \text{'Research'}$

- Select tuples t that belong to relation *EMPLOYEE*
- Compare t to a certain tuple s to find employees working for the department ‘Research’
- Use the existential quantifier

$$\boxed{\exists t \in R (Q(t))}$$

Existing a tuple t of the relation R such that the expression $Q(t)$ is TRUE

Example 12 – TRC

- Find employees (SSN) who work for the department “Research”

$$\{ t.SSN \mid t \in \text{EMPLOYEE} \wedge \\ \exists s \in \text{DEPARTMENT} ($$
$$s.DNAME = \text{'Research'} \wedge \\ s.DNUMBER = t.DNO) \}$$

$Q(s)$

Example 13 - TRC

- For each project in “Stafford”, find the project number, the department number that controls the project and the FNAME of the manager

Example 12 – DRC

- Find employees (SSN) who work for the department “Research”

$$\{ s \mid \exists z ($$
$$\langle p, q, r, s, t, u, v, x, y, z \rangle \in \text{EMPLOYEE} \wedge$$
$$\exists a, b (\langle a, b, c, d \rangle \in \text{DEPARTMENT} \wedge$$
$$a = \text{'Research'} \wedge b = z)) \}$$

Example 13 - DRC

- For each project in “Stafford”, find the project number, the department number that controls the project and the FNAME of the manager

Next lecture

- Data retrieval
 - Advanced query
 - Set operations
 - Nested queries

- Read
 - Chapter 5 [The complete book, 2E]
 - Chapter 6, 8 [Fundamentals of Database System, 7E]

- Watch
 - Video clip

